Class Project Phase III – Part 1 Submission

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1. Architecture explanation and architecture diagram:

For the SunDevil Connect project, I selected the **Model-View-Controller (MVC)** architecture. MVC is a strong fit for this system because the project includes multiple user roles, several system functionalities, and clear interaction between system data, user actions, and presentation. SunDevil Connect allows students to browse and register for events, allows club leaders to create events and post announcements, and supports future administrative oversight. Since these features involve separate responsibilities, MVC provides a clean way to organize the system.

In this architecture, the **Model** represents the core system data and business objects, such as User, Student, ClubLeader, Club, Event, Announcement, and EventRegistration. These classes store the system's state and represent the main entities identified in earlier project phases. The **Controller** layer manages application behavior and user requests. For example, EventController handles browsing and creating events, RegistrationController manages event registration, and ClubController manages club-related functionality such as posting announcements and viewing club pages. The **View** layer is represented in the current implementation through the console-based user interaction in Main, which displays information to the user and demonstrates the system flow.

Furthermore, MVC is beneficial for this project because it supports the **separation of concerns**. Data is stored in model classes, logic is handled by controllers, and presentation is kept separate. This makes the code easier to understand, maintain, test, and extend. As the system grows in subsequent increments, the current console view could also be replaced by a graphical or web user interface without requiring major changes to the model classes.

In my opinion, an alternative architecture that could be considered is the **Client-Server architecture**. Client-Server is useful when the system is distributed between a front-end client and a back-end server. While SunDevil Connect may eventually be deployed in that way, Client-Server alone does not provide the same internal design structure as MVC. It explains where components run, but it does not clearly organize the codebase into data, logic, and presentation responsibilities.

For this reason, MVC is the better choice for the current stage of the project. It directly supports the system's required functionality and gives the project a strong, maintainable foundation for implementation.

2. C4 diagrams:

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2.1 Context Diagram:

The context diagram provides a high-level view of the SunDevil Connect system and its interactions with external users and systems. It shows how students, club leaders, and administrators interact with the system, as well as how the system communicates with external services such as notification handling and data storage.

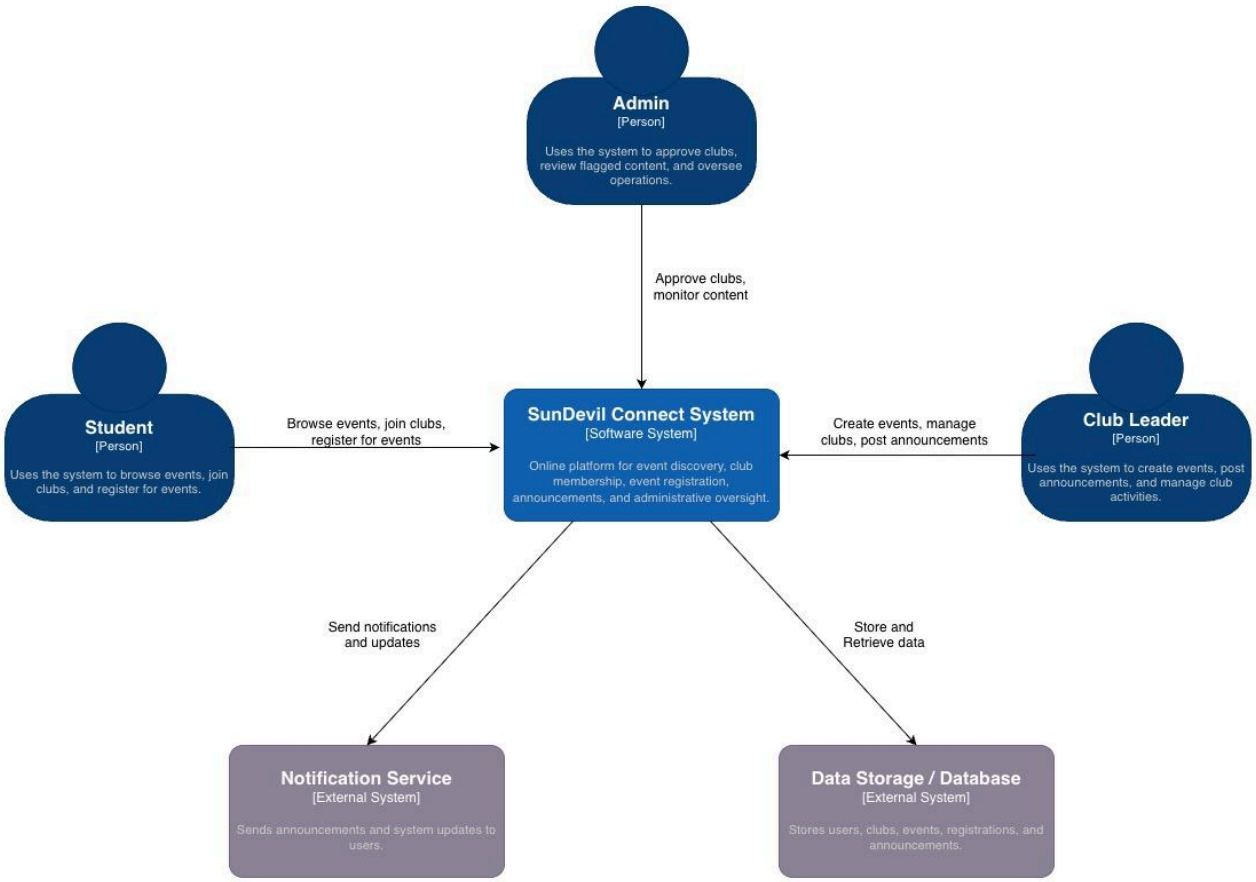

Figure 1: Context Diagram of SunDevil Connect System

2.2 Component Diagram

The component diagram illustrates the internal structure of the SunDevil Connect system. It shows how the system is divided into major components such as Console UI, Event Management, Club Management, Registration Management, Notification Component, Configuration Component, and Data Storage. These components reflect the MVC architecture by separating user interaction, business logic, and data handling.

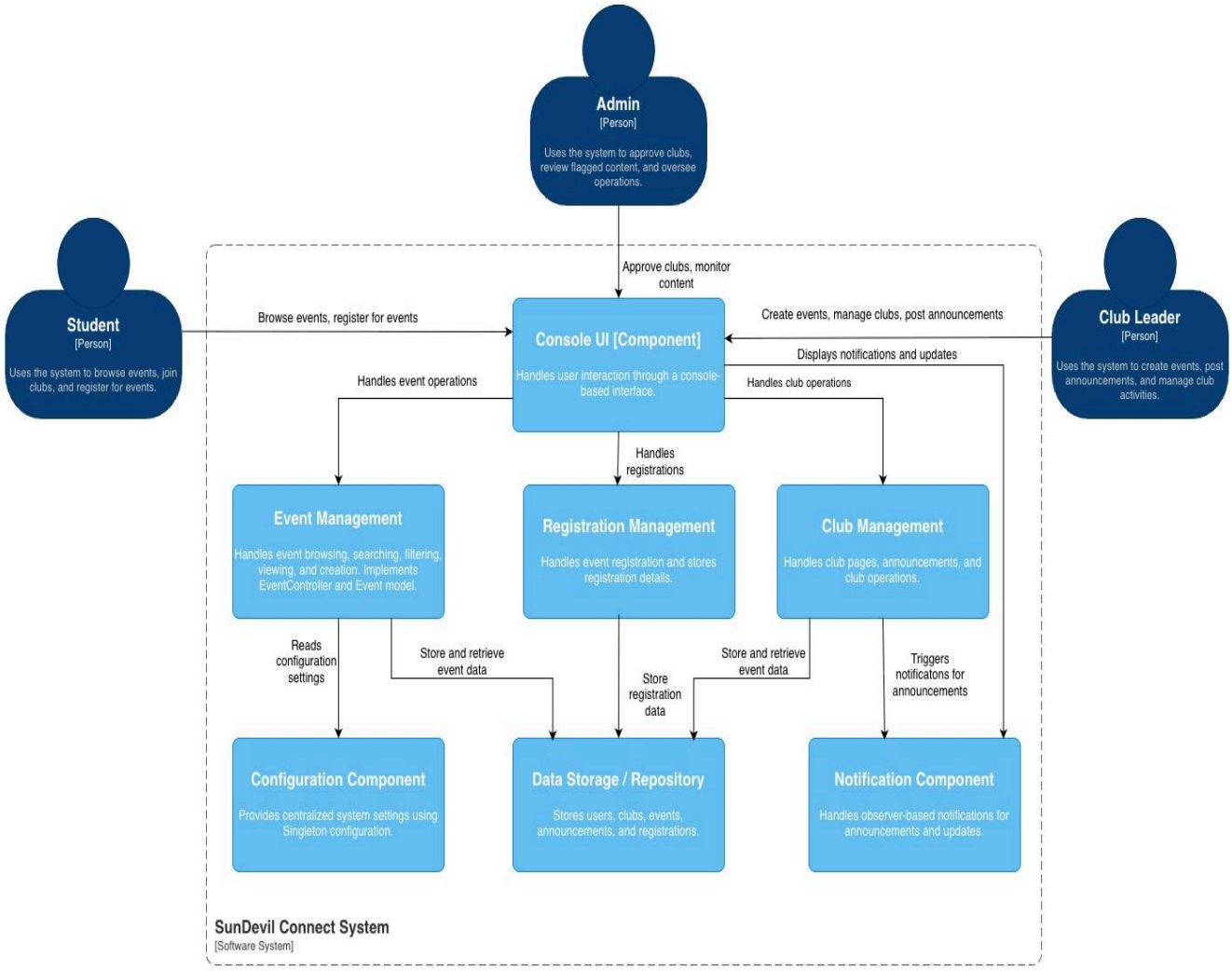

Figure 2: Component Diagram of SunDevil Connect System

2.3 Deployment Diagram (Extra Credit)

The deployment diagram shows how the system is physically executed. In the current implementation, the system runs as a Java console application on a local machine. The Java Runtime Environment (JRE) executes the application, while data is stored in memory during runtime. The diagram also includes the notification service used for system updates.

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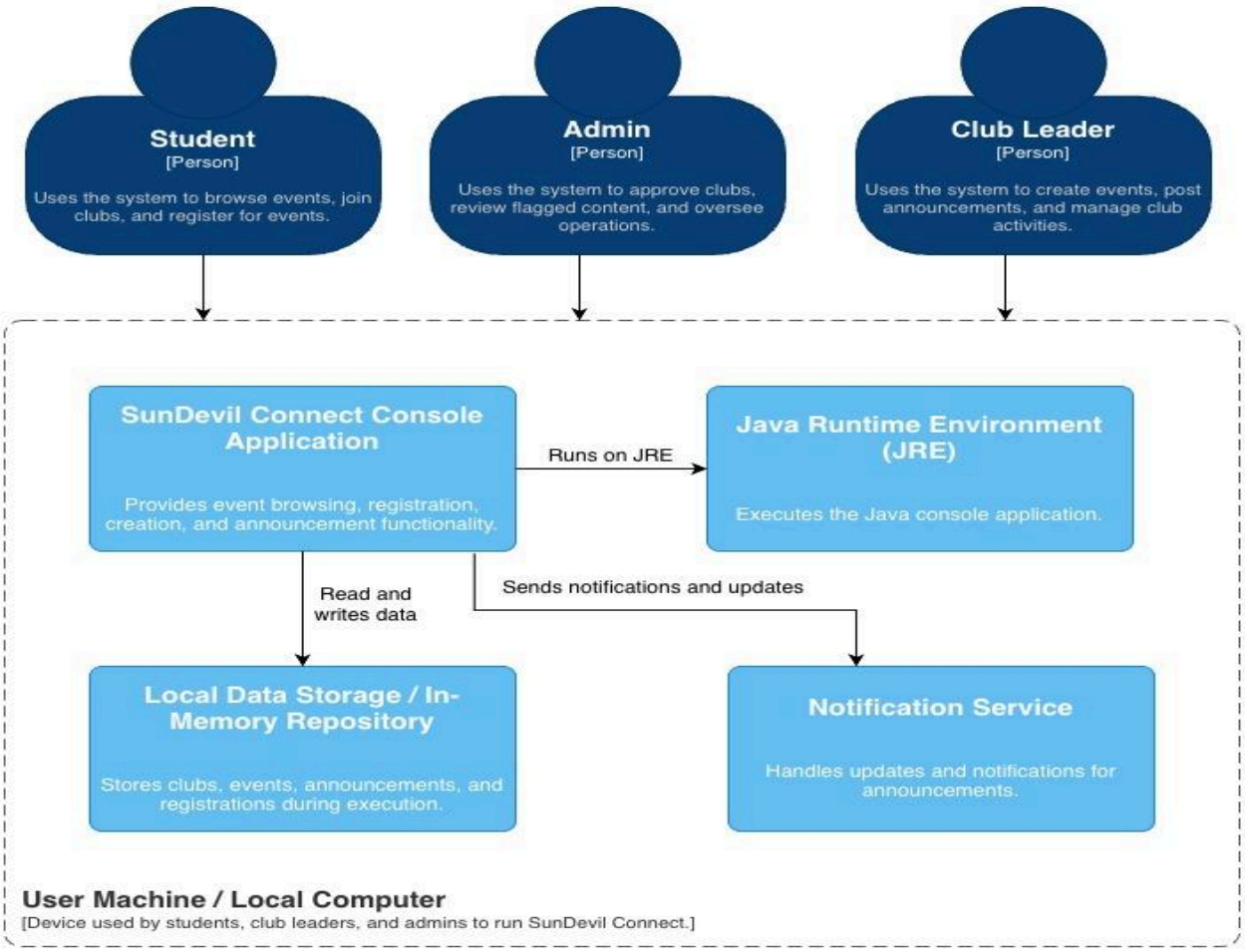

Figure 3: Deployment Diagram of SunDevil Connect System

Phase III Part I code: Please upload the code you have developed for this phase: SunDevilConnect Code

Video Link: Prepare a 3-5 minute video explaining your implementation and the demo phase III Part I implementation.